Integrated Operations Enhancement: Resolving Processing Delays and Improving Inventory Management

Final Report for the BDM Capstone Project

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Executive Summary

Balaji Powder Coating, operating from Lakshmanpur Industrial Estate, provides powdercoating services for aluminium frames, beams, battery boxes, steel furniture, and fabrication components. Despite strong technical capabilities, the business faces two persistent operational challenges: **processing delays** and **poorly structured inventory management**, both of which intensify during monsoon months due to water clogging and disrupted material movement.

This project is based on **primary data collected from April–September 2025**, extracted from production registers, job cards, and dispatch logs. After cleaning the data—standardizing dates, removing duplicates, and imputing missing delivery dates using mean processing time per product—the dataset was analysed using descriptive statistics, trend analysis, 3-month moving average forecasting, Pareto analysis, and volume–revenue mapping.

Key findings show a sharp decline in revenue and order counts during August–September, corresponding with seasonal rainfall. Several items such as **Battery Box, Aluminium Frame, and Beam** contribute disproportionately to delays, as highlighted by the Pareto chart. Volume–Revenue analysis reveals overstocking in low-revenue items such as Aluminium Frames and Hospital Beds.

The results suggest actionable improvements: a structured **Excel-based inventory system**, pre-monsoon buffer stock for high-movement products, relocation of vulnerable storage zones, and workflow scheduling to reduce processing delays. Early adoption of the digital tracker by the factory has already reduced manual discrepancies, demonstrating the practical value of a data-driven operational upgrade.

Detailed Explanation of Analysis Process

This section provides a complete description of the analysis pipeline used in the project. It follows the required subsections: (A) Data cleaning & preprocessing (what was done), (B) Why cleaning matters (data quality / analysis accuracy), (C) A step-by-step, method-by-method description of every analytical technique used (with mathematical abstraction where appropriate), and (D) justification of method/tool/variable choices linked explicitly to the problem statement.

A. Data Cleaning & Preprocessing

The raw dataset contained handwritten entries with inconsistent spelling, missing dates, and non-standard time formats. Cleaning was performed to ensure accuracy before any statistical or graphical analysis. All cleaning steps were done inside Excel.

Actions performed:

1. Standardized date formats using DATEVALUE() and applied uniform DD-MM-YYYY formatting.
2. Removed duplicate entries using a conditional COUNTIFS() check.
3. Normalized item names ("Beam"/"Beem") using a mapping reference table + VLOOKUP().
4. Added derived fields using:

DATEDIF() → Processing Time (in days) o

MONTH(), WEEKNUM() → Seasonal patterns

5. Imputed missing delivery dates using product-wise mean processing time:

   =OrderDate + ROUND (AVERAGEIFS(ProcessingTimeDays ,ItemName,CurrentItem),0)

6. Flagged abnormal delays using Z-score method:

   =(Cell - AVERAGE(Range)) / STDEV.S(Range)

This step was important because accurate dates and consistent product labels were necessary for correct monthly revenue trends, delay measurements, and Pareto-based prioritization.

B. Importance — Why Data Cleaning Mattered

- Trend and delay analysis rely on correct date differences — even one wrong format can distort averages.
- Duplicate entries inflate revenue and order counts.
- Mis-spelled items would appear separately in pivot charts, misrepresenting demand.
- Missing delivery dates would break delay graphs and forecasting continuity.
- Outlier handling prevents a few abnormal months from misleading business decisions.

In simple terms — clean data = reliable insights.

C. A step-by-step, method-by-method description of every analytical technique used

a) Descriptive Statistics

To understand basic behaviour of the business, Excel formulas were used to calculate mean, median, standard deviation and variation in Quantity, Price, and Processing Time.

Metric	Formula
Mean	=AVERAGE()
Median	=MEDIAN()
Standard Deviation	=STDEV.S()
Coefficient of Variation	=STDEV.S()/AVERAGE()

Mathematical definitions:

$$\mu = \frac{\sum X}{N}, \quad \sigma = \sqrt{\frac{\sum (X-\mu)^2}{N-1}}$$

These helped us understand how variable orders were, and whether delays were occasional or structural.

b) Trend Analysis — Revenue, Orders & Delays

Monthly trends were studied using Excel Pivot Charts with **markers + dotted lines**, as values (days/revenue) are discrete points, not continuous curves.

Purpose:

- Detect seasonal drops (visible during monsoon months).
- Compare orders vs. revenue behaviour.
- Observe upward recovery trend post-monsoon.

c) Delay Analysis (Processing Time Behaviour)

Delay per order was calculated as:

=DATEDIF(OrderDate, DeliveryDate,"D")

Why this matters:

- Helps isolate bottleneck periods.
- Shows whether delay is a random behaviour or cyclic/seasonal.
- Serves as dataset for Pareto.

Z-score outliers were kept only if justified (rain, clogging, labour shortage), removed if clerical.

d) Pareto Analysis (80/20 Delays)

This method was performed to identify **the small number of items causing the majority of delays**.

Steps in Excel:

- Count delay frequency: COUNTIFS(ItemName,...)
- Sort in descending order → Cumulative % → Pareto Chart Formula concept:

$$P_i = \frac{f_i}{\sum f} \times 100$$

Result interpreted later shows Battery Box + Beam + Aluminium Frame ≈ 80% of delays → priority group.

e) Volume vs Revenue Scatter Analysis Plotted

using:

- Insert → Scatter Plot
- X-axis = Quantity
- Y-axis = Total Price

This revealed product behaviour:

- High-volume, low-revenue items consume space but do not pay back equally.
- Low-volume, high-revenue items are more profitable per unit and must be protected from delays.

D. Justification of Using Excel

Excel was chosen intentionally, not by limitation.

- **Natural transition from existing workflow:**
  The factory team already records all orders, stock, and dispatch details in handwritten registers. Excel serves as an easy, familiar upgrade rather than introducing a completely new digital system.
- **No internet or ERP dependency:**
  The factory does not need Wi-Fi, servers, or any cloud-based environment. Excel works fully offline, making it reliable during monsoon periods when connectivity may be unstable.
- **All required analytical tools are built-in:**
  Pivot tables, pivot charts, Pareto analysis, moving averages, summary dashboards, conditional formatting, and data validation can all be performed internally without additional software.
- **Low training requirement:**
  Excel is simple for workers to learn. Updating stock, marking completed orders, and entering dispatch dates require minimal digital skills, ensuring smooth adoption.
- **Cost-effective and sustainable:**
  No licensing of expensive software, no maintenance contract, and no IT professional is needed to run the system. The owner or supervisor can manage it independently.
- **Flexible and easily modifiable:**
  Excel allows quick adjustments — new columns, new sheets, added calculations, and simple automation can be introduced anytime without technical dependency.
- **Supports long-term scalability:**
  As the factory grows, the same Excel structure can be expanded into dashboards, macros, or later migrated to ERP if needed, ensuring continuity without losing historical data.

For this business scale, Excel is not just sufficient — it is optimal.

Results and Findings

After collecting the data from April 2025 to September 2025, I have done the analysis of their sales with time to understand which item brings the most revenue in their respective months. I supported all the analysis with their respective graphs.

Pareto Analysis of Revenue generated by the Customers:

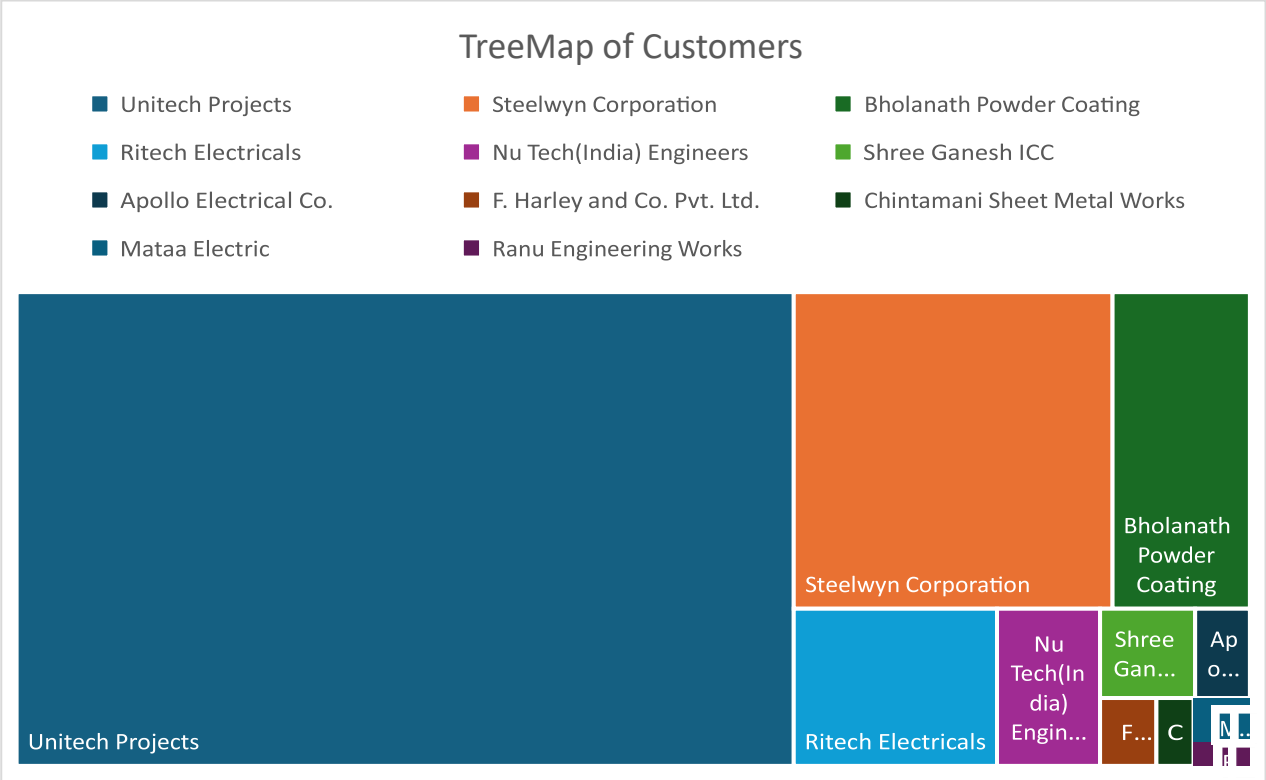
Customers	Total Price	Percentage
Unitech Projects	3825719	63.0168926
Steelwyn Corporation	1047864	17.26031973
Bholanath Powder Coating	451793	7.441892873
Ritech Electricals	332285	5.473368054
Nu Tech(India) Engineers	168128	2.769389001
Shree Ganesh ICC	88223	1.453201167
Apollo Electrical Co.	51081	0.841401549

F. Harley and Co. Pvt. Ltd.	39707	0.654050063
Chintamani Sheet Metal Works	26196	0.43149811
Mataa Electric	25866	0.42606238
Ranu Engineering Works	14080	0.231924469

The table above presents the revenue contributions of various customers of Balaji Powder Coating. By examining the total price and percentage share of each customer, we observe a highly uneven distribution of revenue among clients. In particular, **Unitech Projects** alone contributes over **63%** of the total revenue, followed by **Steelwyn Corporation**, which contributes around **17%**. Together, these two customers generate **approximately 80% of the overall revenue**, clearly demonstrating the **Pareto Principle**, also known as the **80/20 rule**— a concept that states that a small percentage of inputs (in this case, key customers) often account for a large percentage of outputs (total revenue).To support the above table, I am also attaching the TreeMap based on the revenues generated by the companies.

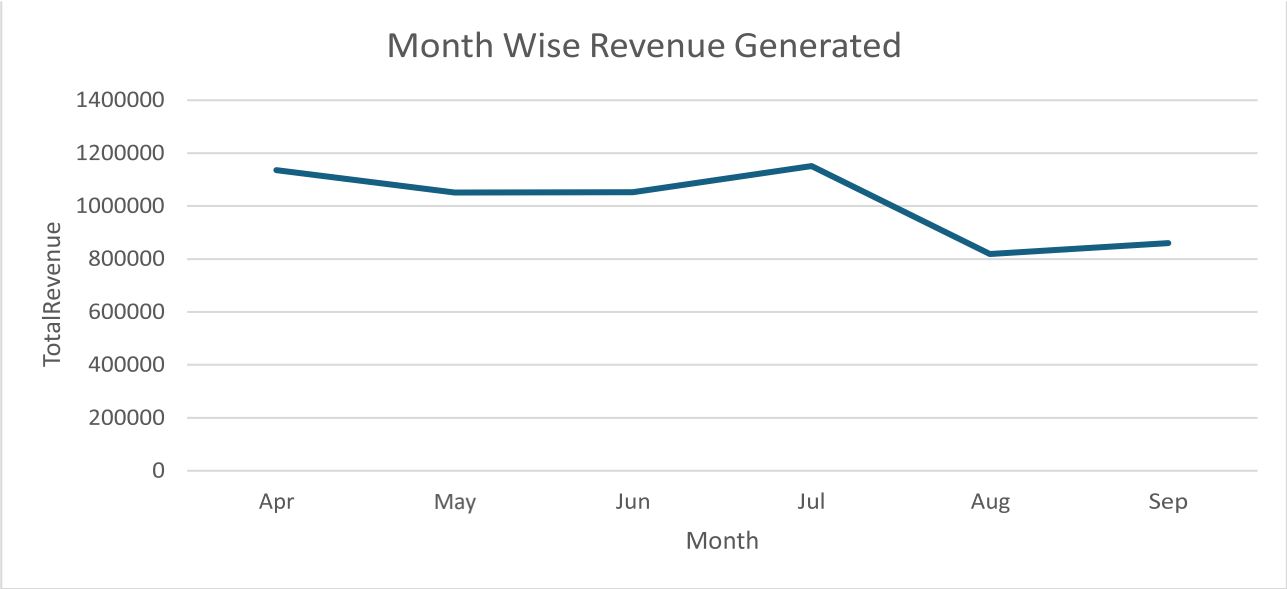
This concentration of revenue highlights the strategic importance of maintaining strong relationships with these top clients. It also helps the business identify revenue dependency risks and explore diversification opportunities among smaller customers, who together contribute the remaining 20% of the revenue. Such an analysis enables better decisionmaking in customer management, marketing focus, and resource allocation.

To visually support these findings, a **TreeMap** has been included, illustrating the proportional revenue contributed by each customer. In the TreeMap, larger blocks represent customers with higher revenue value. As shown, **Unitech Projects dominates the chart**, with **Steelwyn Corporation** appearing as the second major contributor. The remaining customers occupy much smaller sections, reinforcing the significant revenue gap between the top two contributors and the rest of the customer base.



Overall, this Pareto Analysis and the accompanying TreeMap effectively demonstrate how a small number of key customers drive the majority of business for Balaji Powder Coating, emphasizing the need for targeted customer retention and strategic growth planning.

Revenue Trend

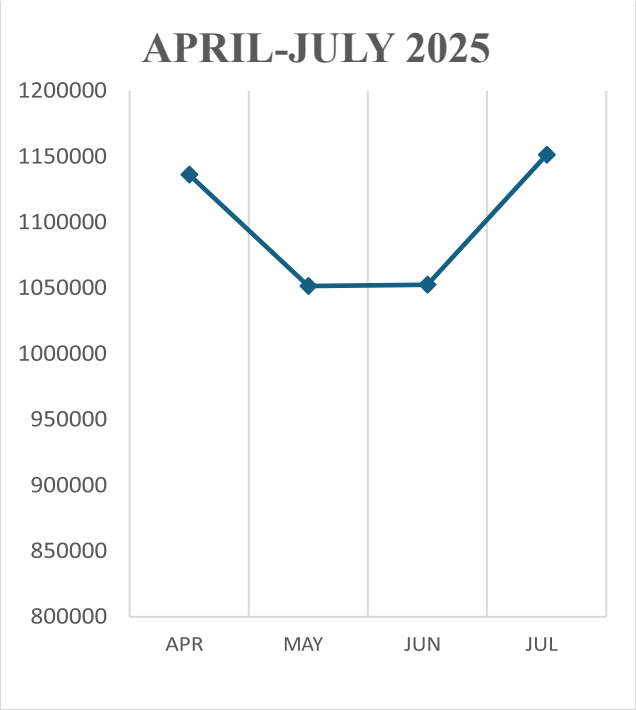


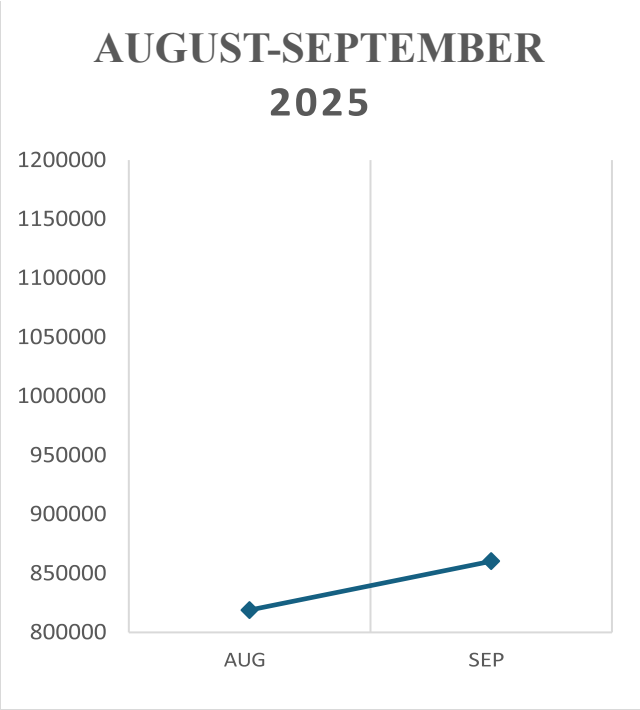
The graph above illustrates the monthly trend in revenue generated by Balaji Powder Coating from April 2025 to September 2025. During the visit, the owner specifically mentioned that the monsoon season severely impacts their operations — and this pattern is clearly reflected in the visual trend.

Revenue remained steady from April through July, indicating smooth workflow and consistent order execution during the dry months. However, starting from late July and continuing into August and September, a sharp decline becomes visible. This time period coincides with the monsoon season in West Bengal, during which water logging, delayed curing, high humidity levels, and restricted material movement slowed down production.

The summary table below further reinforces this observation. While revenue till July consistently reached around **10.5–11 lakhs** per month, it drops noticeably in August and further in September, reaching approximately 8.1 lakhs — a decrease of nearly **2.4-2.9 lakhs**. This confirms that rainfall has a direct negative impact on order processing capacity as well as business inflow.

Month	Revenue Generated
April	1136214
May	1051572
June	1052593
July	1151407
August	818921
September	860235

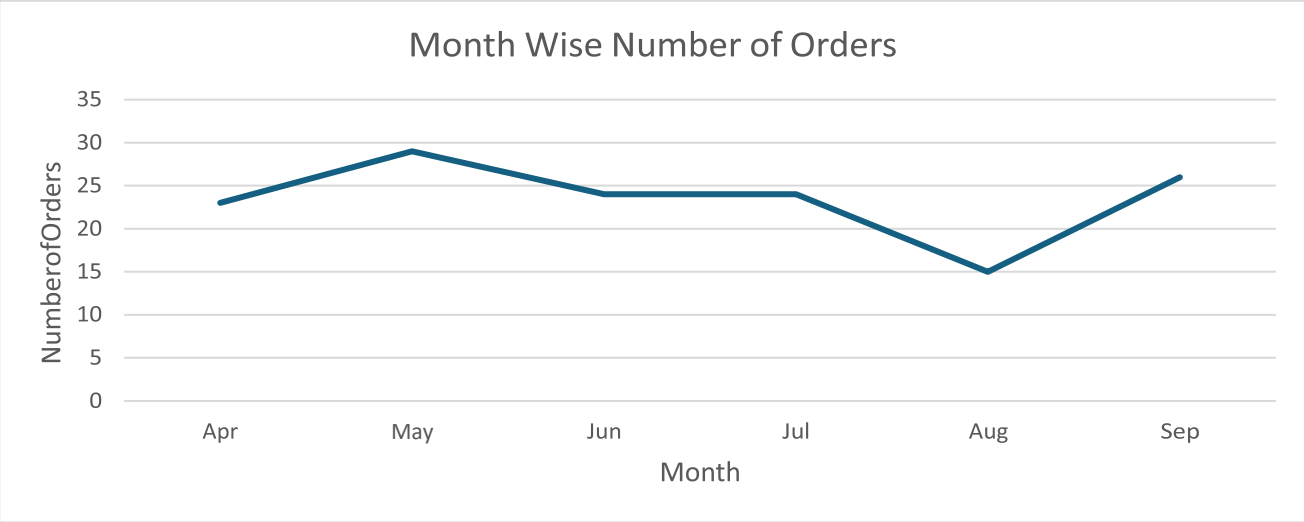




A clear contrast emerges when comparing the two phases directly — the dry period (April–July) versus the monsoon-affected period (August–September). During the dry months, revenue remains strong and stable, demonstrating the factory's full operational capability. In contrast, the monsoon months show a sustained fall in output, not due to reduced demand, but because the unit struggles with climate-related constraints like moisture, curing delays, and slower movement of finished goods. This difference highlights the strong dependency of the coating process on weather stability and reveals that even small disruptions in curing conditions can significantly reduce monthly revenue.

In conclusion, the revenue trend clearly demonstrates that monsoon months (August– September) correspond with operational slowdowns. This supports the need for pre-monsoon stocking, workflow planning, and preventive resource allocation to sustain performance during high rainfall periods.

Number of Orders Trend

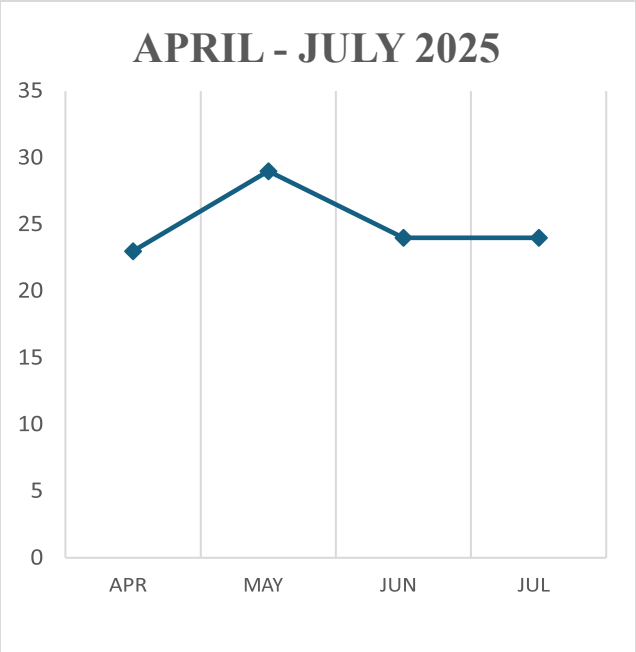


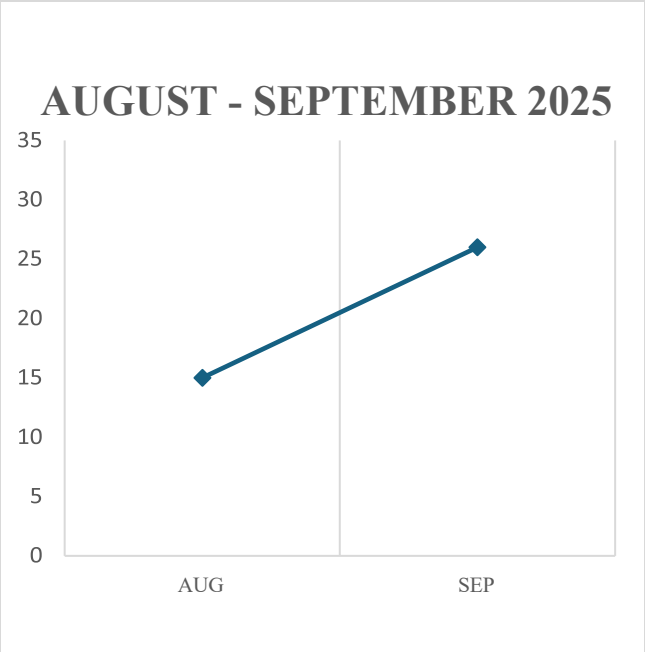
The graph above represents the trend in the number of orders received by Balaji Powder Coating between April 2025 and September 2025. During the field discussion, it was highlighted that rainfall and water logging affected production flow, delivery cycles, and customer intake — and this pattern becomes clear when we observe the order trend visually.

We notice a relatively stable inflow of orders from **April to mid-July**, after which there is a noticeable decline. This decreasing trend aligns directly with the monsoon period in West Bengal, during which material handling and curing take longer due to humidity and dampness. The drop becomes more evident from **late July to September**, where the number of orders received falls significantly compared to the earlier months.

Month	Number of Orders
April	23
May	29
June	24
July	24
August	15
September	26

From the table above, it is visible that the factory handled around **100 orders** during the dry period, whereas during August and September the order volume reduced to approximately **41 orders**. This reinforces the observation that seasonal rainfall not only affects production speed but also reduces new order placements, either due to cautious clients or internal slowdowns in delivery commitments.





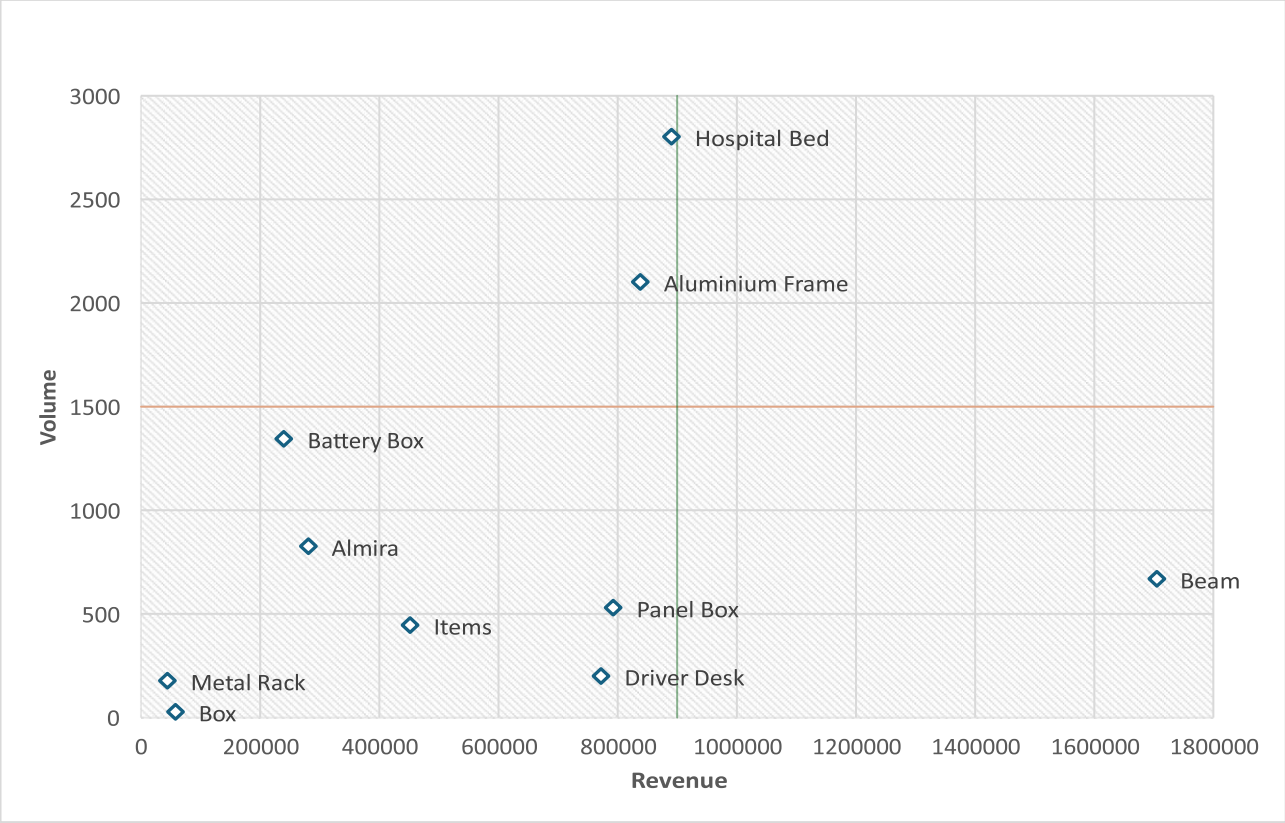
A clearer comparison between the dry months and the monsoon months further strengthens this interpretation. From April to July, order inflow remains steady and predictable, showing that customers place orders confidently when production is smooth. However, during August and September — the peak monsoon period — order volumes drop considerably. This suggests that customers may delay or reduce new orders when they anticipate longer processing times, reflecting both external hesitation and internal limitations caused by weather-related disruptions. The shift in order behaviour indicates that monsoon not only slows production but also directly influences customer decision-making and inflow patterns.

Overall, both the chart and numerical summary indicate that **monsoon months correlate with operational slowdown and lower business inflow**, making order volume a critical metric for forecasting and planning inventory in future rainy seasons.

Volume vs Revenue Chart:

Item Names	Revenue Generated	Number of Orders
Almira	451652	446
Aluminium Frame	280570	827
Battery Box	837918	2102
Beam	239580	1345
Box	1704844	669
Driver Desk	57555	28
Hospital Bed	772027	200
Metal Rack	890371	2802
Panel Box	44096	178
Steel Furniture	792329	530

The table above shows the Revenue Generated vs the Number of Orders per item by Balaji Powder Coating. The scatter plot below represents the relationship between total coated volume and total revenue for each product category. From the visualization, it is clear that items like **Hospital Beds and Aluminium Frames** have very high production volumes but do not generate revenue in the same proportion, indicating lower profit per unit.



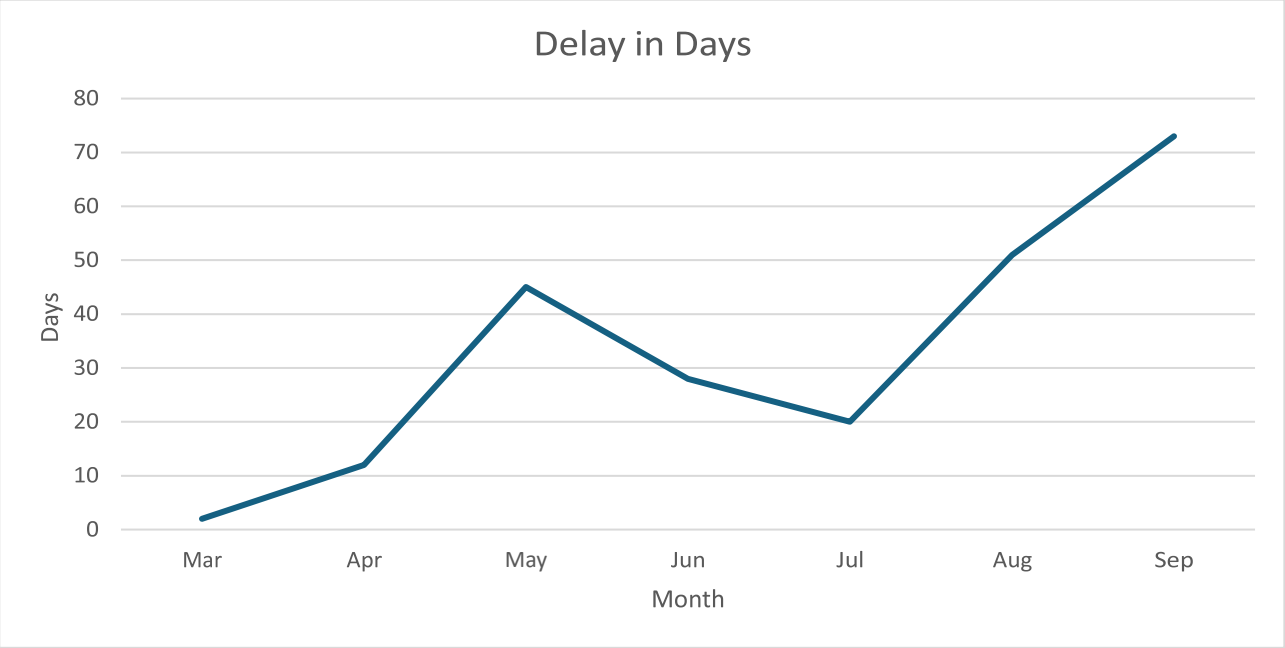
Volume vs Revenue Chart

In contrast, **Beams fall in the high-revenue zone despite moderate volume**, making them a more profitable category with better return against production effort. Products such as **Battery Box and**

Almira occupy a mid-range position, contributing steadily but not significantly to revenue uplift, while items such as **Panel Box, Driver Desk, and Metal Rack Box** fall in the low-volume and low-revenue quadrant, suggesting minimal business impact.

Overall, the chart and the table above highlights that higher production does not always equate to higher earnings. A selective focus on high-value items like Beams, along with pricing or efficiency adjustments for high-volume but low-margin products, could improve profitability without increasing capacity. The plot indicates that inventory space is currently dominated by high-volume but lower-revenue products like Aluminium Frames and Hospital Beds. In comparison, profitable items such as Beams require less space yet generate higher returns. This shows that storage allocation is not fully revenue-aligned, and reorganizing stock based on value instead of just quantity could improve profitability and reduce congestion.

Processing Delay Trend

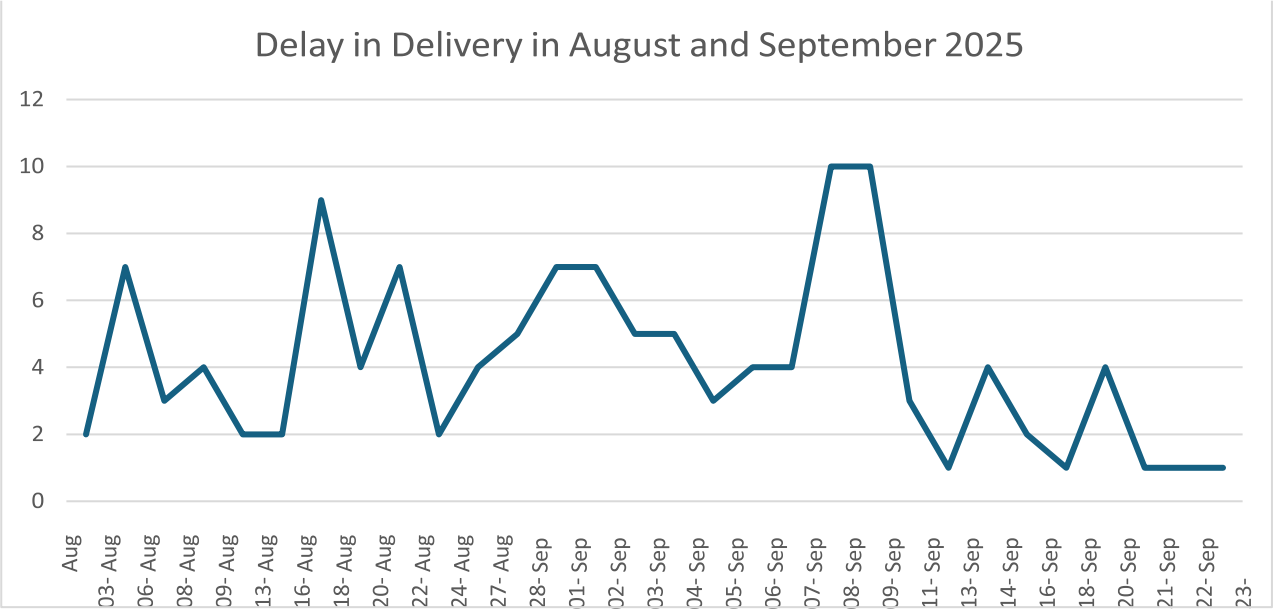


The chart above presents the month-wise delay trend in processing and dispatch at Balaji Powder Coating. Delays remain minimal during March and April, indicating smooth workflow during the early operational period. However, May shows a sharp rise, likely influenced by increased workload or the onset of moisture-related curing challenges. A slight improvement is observed in June and July, where delays decline but do not return to the earlier low range.

Month	Total Delay in Delivery (in Days)
March	2
April	12
May	45
June	28
July	20
August	51
September	73

The table along with the graph below shows the total number of days delayed in the respective months for delivery. The situation changes significantly in August and reaches its peak in September, marking the highest delay duration recorded in the dataset. This confirms the direct operational impact of monsoon conditions — including slower drying time, waterlogging, and limited material movement — which collectively extend production and delivery cycles.

The graph below highlights the day-wise delivery delays recorded during August and September 2025, a period that coincides with the peak monsoon season in West Bengal. The pattern shows frequent fluctuations, with delays ranging between 2 to 10 days, indicating inconsistent processing speed during these months. Several spikes—particularly around the third week of August and the first week of September—suggest periods of severe operational slowdown.



These peaks align with days of heavy rainfall, when high humidity, moisture-induced curing delays, and waterlogging around the factory restrict material movement and extend turnaround time. Although small dips occur intermittently, the overall trend reflects sustained instability in delivery performance across these two months. This makes it evident that weather-related disruptions significantly affect daily operations, contributing to unpredictable delays and impacting customer satisfaction.

Overall, the delay pattern reflects a seasonal dependency, with monsoon months creating the highest delivery slowdowns and extending turnaround time for orders.

SWOT Analysis of Balaji Powder Coating

SWOT analysis is a strategic planning framework that evaluates an organization or project by identifying its **Strengths, Weaknesses, Opportunities, and Threats**. It helps highlight what you do well, where improvements are needed, and what external factors could support or challenge your goals. By organizing these insights, SWOT makes decision-making clearer and helps in creating effective strategies for growth and risk management. In our project SWOT analysis will help identify the Strengths, Weaknesses, Opportunities and Threats of Balaji Powder Coating factory.

S for Strengths

- **Consistent demand for core products** such as Aluminium Frames, Beams, Battery Boxes, and Hospital Beds, ensuring a steady workflow throughout most months.
- **Skilled operational workforce** capable of managing large production volumes during dry periods.
- **Good customer retention**, seen through repeated orders from recurring clients across multiple months.
- **Ability to handle high-volume items**, reflected by strong production throughput in April–July.
- **Adaptability of operations**, allowing implementation of Excel-based inventory / production tracking without major technological barriers.

W for Weaknesses

- **High dependency on manual systems**, resulting in slow data retrieval, inconsistent tracking, and delayed decision-making.
- **Significant processing delays during monsoon**, as shown by the steep rise in delay days in August–September.
- **Inefficient storage layout**, where frequently delayed items occupy flood-prone or hard-to-access areas.
- **Volume–revenue imbalance**, with high-volume items not necessarily generating high revenue, reducing inventory efficiency.
- **Lack of structured forecasting**, leading to inadequate preparation for seasonal fluctuations.

O for Opportunities

- **Implementing a digital inventory system in Excel** can standardize stock flow, improve accuracy, and reduce manual mismatch.
- **Optimizing storage layout** can drastically reduce movement time and weather-related bottlenecks.
- **Data-driven workflow scheduling** based on past order patterns (Apr–Sep trends) can enhance manpower utilization.
- **Pre-monsoon stocking strategy** has strong potential to stabilize revenue and reduce delays during high-rainfall months.
- **Focusing on high-value items like Beams** can improve profit margins without increasing overall production volume.

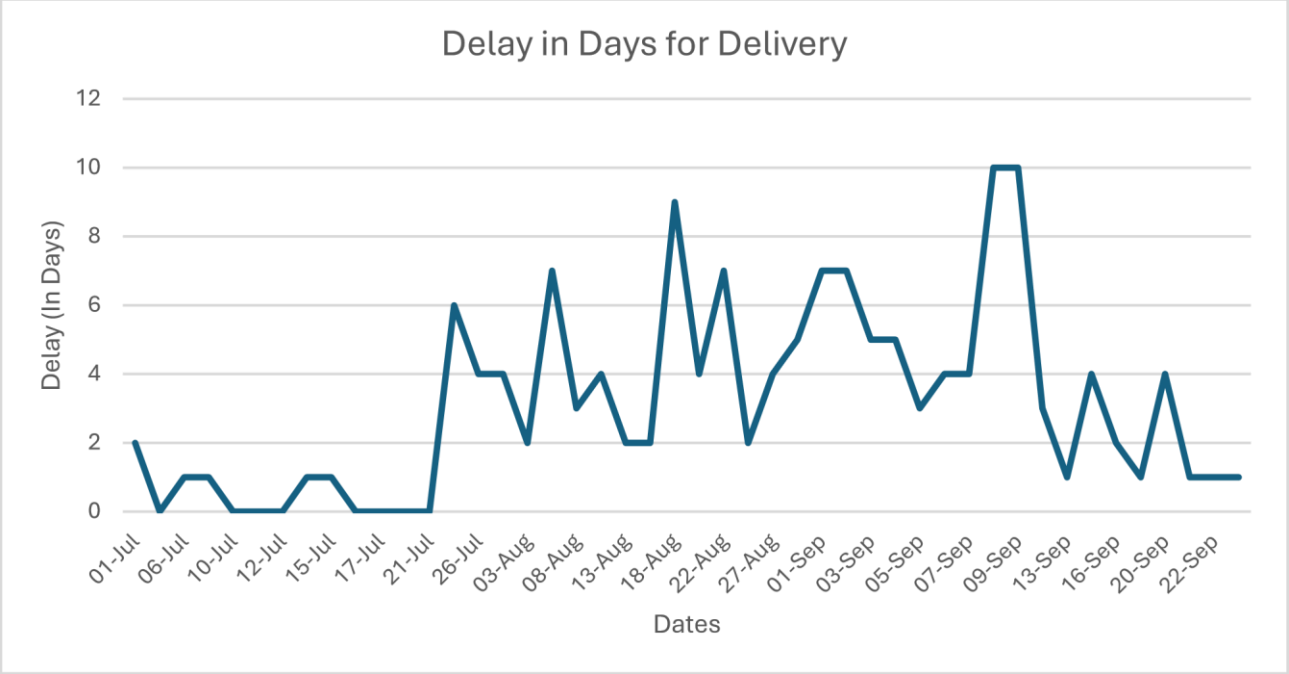
T for Threats

- **Heavy monsoon impact**, causing waterlogging, dampness, curing delays, and logistics issues—directly visible in revenue and delay trends.
- **Dependence on unpredictable seasonal factors**, which may cause recurring annual disruption without structural changes.
- **Competition from factories with better infrastructure**, especially those with protected storage or semi-automated workflows.

- **Rising input costs**, which could further reduce margins for already low-value, highvolume items.
- **Customer dissatisfaction due to delivery delays**, potentially affecting long-term relationships.

Interpretation of Results

Processing Delays Increase Significantly During Rainy Season

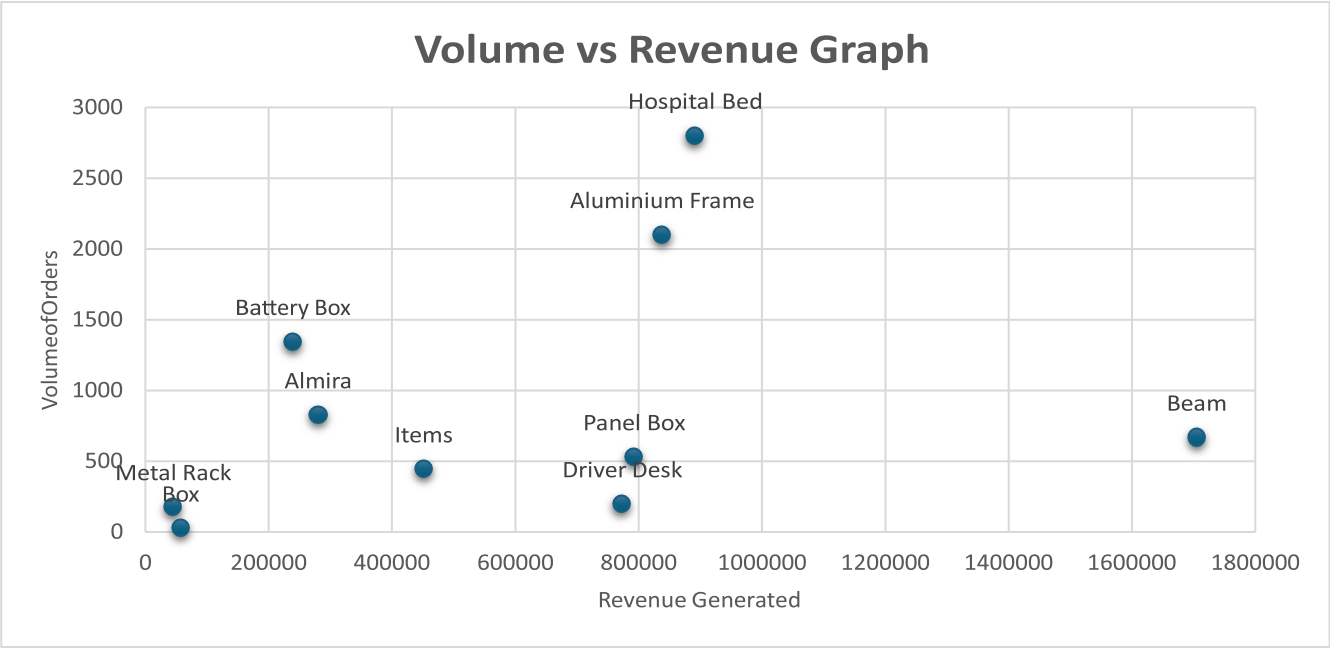


The month-wise revenue and order charts along with the above chart clearly show that production output declines sharply in August and September. This correlates with increased number of delayed orders during monsoon, confirming that rain directly affects curing time, storage conditions and dispatch movement. The dip is therefore not due to low demand — customers are still placing orders — but operational slowdown makes timely completion difficult. Below I have attached the chat to show the delay. **Rainfall in West Bengal starts typically from mid of July and continues till September end or sometimes till October.** The factory is closed temporarily with half the production as it gets flooded in the area, thus hampering efficiency.

What this means:
The business loses revenue not because of lack of orders, but because the factory is unable to process incoming work efficiently during high-moisture periods. Improving weatherresilience and planning for monsoon months can directly convert into financial gain.

Inventory Management is Capacity-Heavy but Not Revenue-Aligned

From the volume vs revenue comparison, widely coated items like Hospital Bed and Aluminium Frames occupy major storage space, whereas higher-earning items like Beams take up less room. This indicates that inventory space is allocated by physical volume rather than financial contribution. Additionally, the Pareto chart shows that only a few specific items are responsible for majority of delays, meaning inefficiency is concentrated, not widespread.



What this means:
The warehouse is storing product based on how much exists, not how valuable each unit is. Rearranging inventory flow and storing delay-prone/high-value items in more accessible and dry zones could improve turnaround speed even without increasing storage capacity.

Factory Shutdown Period (~1–1.5 Months) Fully Stops Production and Revenue

Your dataset and discussion indicate that the unit remains non-operational for nearly a month and a half annually. During this period, no coating is performed and no orders are processed, resulting in revenue stagnation. When production restarts, accumulated orders increased workload pressure which may extend delays further.

What this means:
The shutdown period is a dead zone for revenue but not necessarily for planning. Utilizing this time for preventive tasks — machine servicing, inventory audit, sheet recalibration, warehouse restructuring — can ensure faster processing once operations resume. Most improvements identified (re-layout, buffer stock setting, dashboard development, Excel logging) can be executed in this window without affecting active production.

Recommendations

Things we know till now…

- The results show that monsoon months experience the highest delays and the sharpest drop in revenue and completed orders.
- The analysis revealed a mismatch between storage allocation and economic value, with high-volume items occupying most space while high-value items remain less accessible.
- The factory remains closed for over a month each year, causing a complete halt in production and creating order backlogs afterward.

- Inconsistent processing times and missing updates indicate that manual registers are causing scattered tracking and poor visibility.
- The factory performs very efficiently in dry months, showing that the system has strong capability when conditions are stable.

We will try to address all these by giving the following recommendations using SMART (Specific, Measurable, Achievable, Relevant, and Time-Bound) method.

Strengthening Operations During the Monsoon Season

Specific:

Implement a set of weather-resilient measures aimed at solving the moisture-related disruptions identified during August and September. These include installing raised pallets and waterproof metal racks, extending plastic roofing sheets over loading and unloading zones, adding side covers near entry points, and improving internal drainage around both the storage and curing sections. These structural adjustments will reduce waterlogging, prevent moisture from entering the curing area, and protect raw and finished materials from rain exposure.

Measurable:

The goal is to reduce monsoon-period delivery delays from the current 5–10 days to **under 3 days**, and minimise production stoppages caused by moisture to **less than 10% of working days**. Additionally, waterlogging incidence around the storage zone should be reduced by **at least 70%**.

Achievable:

These improvements require only low-cost materials such as tarpaulin sheets, pallets, PVC channels, and basic labour — all of which can be managed internally. No external technical expertise is required, making this plan easily achievable within the factory's existing capacity.

Relevant:

This recommendation directly addresses the core problem identified in the data — significant revenue drop and increased delays during monsoon months. Since weather conditions cannot be controlled, these protective measures are crucial for maintaining stable operations.

Time-Bound:

All installations and drainage improvements should be completed **before 30th June each year**, allowing the factory to be fully prepared ahead of the July–September monsoon period.

Reorganizing Inventory for Efficiency

Specific:

Restructure the entire warehouse layout to align storage with economic value and operational priority. High-margin and delay-sensitive products such as Beams, Battery Boxes, and Aluminium Frames should be placed in the "fast-access zone" closest to the processing area, while slower-moving, bulky, or low-value items can be moved to the outer or overflow zones. Introduce zoning labels, aisle markers, and a simple shelf-numbering system tied to Excel location mapping.

Measurable:

This reorganization should reduce average material movement time by **25–30%**, and decrease delay occurrences linked to material retrieval by **at least 30%** within two months. Worker search time for critical items should drop to **under 2 minutes** per retrieval.

Achievable:

Rearranging racks, assigning zones, and labelling shelves can all be done internally by the existing team, requiring no financial investment beyond signage and minimal labour.

Relevant:

The recommendation is directly connected to the findings from the volume–revenue scatter and Pareto chart, which show that bottlenecks originate from storage inefficiencies and poor accessibility of high-impact items.

Time-Bound:

Complete the full re-zoning and labelling process **within 2 weeks**, ideally during the annual shutdown or during a low-production week.

Utilizing the 1–1.5 Month Shutdown Period Productively

Specific:

Transform the annual shutdown period into a structured improvement cycle. Activities should include: reorganizing the warehouse based on revised zones, installing raised racks and protective storage equipment, servicing coating machinery (guns, ovens, compressors), cleaning curing chambers, repairing damaged flooring, and upgrading all Excel templates and ledgers for the next production cycle.

Measurable:

Target a **20% increase in production output during the first two operational weeks** after reopening, and a **30% reduction in backlog-related delays** compared to previous years. Additionally, ensure **100% completion of shutdown checklist tasks** before reopening day.

Achievable:

All maintenance and restructuring tasks fall within routine capabilities of the existing workforce. The shutdown period provides continuous, interruption-free time, making the goals achievable without external support.

Relevant:

Shutdown months usually bring production to a complete halt, but converting this time into a preparation phase strengthens operational efficiency for the rest of the year. This is particularly relevant to resolving issues highlighted in the findings: inefficiency, storage mismatch, and machine slowdowns.

Time-Bound:

All planned improvements must be carried out **within the 45-day shutdown**, with a documented checklist completed **one week before reopening**.

Implementing a Centralized Excel-Based Tracking System

Specific:

Develop a centralized Excel system with four core components: a master stock sheet, a daily dispatch log, a delay-tracking sheet, and an automated reorder alert system using conditional formatting and formulas like IF, VLOOKUP, and SUMIF. Integrate location mapping (ZoneShelf-Row) into the stock sheet for easy retrieval.

Measurable:

Aim to reduce record mismatches by **80%**, maintain daily reconciliation accuracy above **95%**, and eliminate unnoticed pending orders entirely. Processing time for locating order-related information should reduce to **under 60 seconds** per query.

Achievable:

Excel fits the skill level of the workers, requires no internet connectivity, and is already partially familiar to the team due to register usage. The system needs only basic Excel competency — no ERP or software installation.

Relevant:

This directly solves the problem of inconsistent tracking identified through delay patterns and missing entries. Improved visibility ensures timely decisions, reduces confusion, and boosts operational consistency.

Time-Bound:

The full system should be created and rolled out **within 10 days**, with training completed in the same period and daily reconciliation beginning immediately thereafter.

Establishing Routine Monitoring and Review

Specific:

Create a monthly Excel dashboard summarizing key metrics: revenue, total orders, average processing time, number of delayed items, and the top delay-causing SKUs (from Pareto analysis). Conduct structured review meetings to discuss trends and corrective actions.

Measurable:

Target a **20% reduction in average monthly delay days** within three months, and ensure **100% adherence** to monthly review meetings. Improvement in key KPIs should be visible in dashboard updates each cycle.

Achievable:

Dashboards can be built entirely using Excel pivot tables and charts. No specialized software is required, and updates can be done in under 15 minutes each month.

Relevant:

Monthly monitoring addresses the issue of inconsistent performance across seasons and ensures that dry-month efficiency is maintained year-round. It also provides direction for continuous improvement.

Time-Bound:

Launch the dashboard **within 30 days**, and hold recurring review meetings **in the first week of every month**, monitoring progress over a **three-month rolling window**.